

STAT 412 HW07 — Question 9

Problem

A marketing expert for a pasta-making company believes that 40% of pasta lovers prefer lasagna. If 12 out of 20 pasta lovers choose lasagna over other pastas, what can be concluded about the expert's claim? Use a 0.05 level of significance.

The conclusion choices include:

- A. Reject H_0 and conclude that there is sufficient evidence that the percentage is not 40%.
- B. Do not reject H_0 and conclude that there is not sufficient evidence that the percentage is not 40%.
- C. Do not reject H_0 and conclude that there is sufficient evidence that the percentage is not 40%.
- D. Reject H_0 and conclude that there is not sufficient evidence that the percentage is not 40%.

Solution

The expert's claim is an equality claim about the proportion:

$$H_0 : p = 0.4.$$

Because the question asks what can be concluded about the claim, without saying specifically above or below, use a two-sided alternative:

$$H_a : p \neq 0.4.$$

This is choice C.

Let X be the number of pasta lovers in the sample who choose lasagna. Under the null hypothesis,

$$X \sim \text{Binomial}(n = 20, p = 0.4).$$

So the platform entries for the binomial model are $p = 0.4$ and $n = 20$.

The observed value is $x = 12$. MyStatLab's feedback uses the doubled one-tail exact-binomial rule for a two-sided test. Since $x = 12$ is greater than $np_0 = 8$,

$$P\text{-value} = 2P(X \geq 12) = 2(0.056526367) \approx 0.1131.$$

Since $0.1131 > 0.05$, do not reject H_0 . The correct conclusion choice is B.

Final Answer

Choice C: $H_0 : p = 0.4$; $H_a : p \neq 0.4$. Test statistic/model: $X \sim \text{Binomial}(n = 20, p = 0.4)$. P-value ≈ 0.1131 ; do not reject H_0 at $\alpha = 0.05$. Conclusion choice B.